



Antipeptide IP-LC detection of LRRK2 and pRab10 in genetic PD

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Summary

PPMI CSF samples were analyzed in 2 batches. In the first batch LRRK2 was measured while in batch 2, the assay was upgraded to a 2-plex that contained both LRRK2 and pRab10. LRRK2 (batch 1 and 2) and pRab10 (batch 2) were measured using an immunoprecipitation (IP) liquid-chromatography mass spectrometry (LC-MS) assay developed at Biogen. The assay utilizes the SISCAPA (stable isotope standards and capture by antipeptide antibodies) approach where samples are digested using a protease (trypsin) and then antibodies are used to isolate peptides with unique sequences. Isotopically labeled “heavy” peptide with the same sequence are spiked into the samples at a known concentration. Assay is an updated method of a previously published assay (Mabrouk et al., 2020)

Use Note: The data described in this Methods document supersedes Project 221 data downloaded prior to May 1, 2025, due to the application of a batch correction procedure to normalize results for analytical purposes (see below).

Method

Batch 1 (2022)

Sample Preparation and Digestion

2 aliquots of 250 ul CSF samples were pooled into a deep well 96 well plate (1ml). 50 ul of 10X RIPA buffer (Sigma # R0278) was added to each sample. 20 ug of TPCK treated trypsin (Worthington LS003741) in PBS was added to each sample. Samples were then put on a thermomixer at 1400 RPM at 40C for 90 minutes. Following trypsin digestion samples were put on ice for 5 min.

Immunoprecipitation

2 pgs of heavy peptide for LRRK2 (batch 1 and 2) and pRab10 (batch 2) were added to each sample.

For batch 1 LRRK2 (Neuromab N241A) mAb was conjugated to streptavidin cartridges compatible with the Agilent AssayMap Bravo liquid handler platform. Digested sample was run over the antibody conjugated cartridges to IP peptides.





Wash and Elution

Peptides were eluted off the streptavidin-antibody cartridges with with 50 ul H₂O + 0.1% formic acid and 5% acetonitrile.

LC-MS analysis

Samples were injected onto an Evosep One LC system running in Whisper mode (40 SPD) Evosep One was coupled to an Easy nano source and Thermo QE HFX orbitrap mass spectrometer operating in PRM mode. KAEEGDLLVNPQPR was detected as a +3 ion (m/z 560.9566). FHTITTSYYR was also detected in a +3 charge state (456.8710). Peak identification and quantification were performed in Skyline.

Batch 2 (2024)

Sample Preparation and Digestion

2 aliquots of 250 ul CSF samples were pooled into a deep well 96 well plate (1ml). 50 ul of 10X RIPA buffer (Sigma # R0278) was added to each sample. 20 ug of TPCK treated trypsin (Worthington LS003741) in PBS was added to each sample. Samples were then put on a thermomixer at 1400 RPM at 40C for 90 minutes. Following trypsin digestion samples were put on ice for 5 min.

Immunoprecipitation

2 pgs of heavy peptide for LRRK2 (batch 1 and 2) and pRab10 (batch 2) were added to each sample.

For batch 1 LRRK2 (Neuromab N241A) mAb was conjugated to streptavidin cartridges compatible with the Agilent AssayMap Bravo liquid handler platform. For batch 2, the process was same processed was used as above except the MJFR20 (pRab) antibody was also biotinylated and conjugated to streptavidin cartridges. Digested sample was run over the antibody conjugated cartridges to IP peptides.

Wash and Elution

Peptides were eluted off the streptavidin-antibody cartridges with with 50 ul H₂O + 0.1% formic acid and 5% acetonitrile.

LC-MS analysis

Samples were injected onto an Evosep One LC system running in Whisper mode (40 SPD) Evosep One was coupled to an Easy nano source and Thermo QE HFX orbitrap mass spectrometer operating in PRM mode. KAEEGDLLVNPQPR was detected as a +3 ion (m/z





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Batch Correction and Normalization

Samples were tested across 2 batches (Batch 1 in 2022 and Batch 2 in 2024). Batch 1 and 2 both contained LRRK2 measurements and a batch effect was observed. The batch effects were estimated based on a linear mixed model, with fixed effects for sex, age and random effects for subjects and rundate (batch). The batch-corrected values are based on the same model but with the rundate effect removed. pRab10 was only measured in batch 2 and therefore no correction factor was applied to those measures.

References

1. Mabrouk OS, Chen S, Edwards AL, Yang M, Hirst WD, Graham DL (2020) Quantitative measurements of LRRK2 in human cerebrospinal fluid demonstrates increased levels in G2019S patients. *Front Neurosci* LRRK2 special edition 14:526.

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